

Cadenza-Camerata Research Concert
Stoller Hall, 29th October 2026

Welcome to the Cadenza Camerata Research Concert

This pioneering event is a collaboration between the Cadenza Project and Manchester Camerata. Your answers will help us understand how to improve music for people with hearing loss and shape future research. Participating in the research activities is entirely voluntary, however.

You will hear the orchestra playing movements of Beethoven's 6th Symphony.

Schedule:

1) Complete the background questionnaire below.
2) Movement I – just sit back and enjoy.
3) Research Activity featuring 10 extracts from movements I and IV played in different ways. You will give feedback by the questionnaire contained within this booklet.
4) Movement V –sit back and enjoy or do the optional research task
5) Q&A with members of the Cadenza Team on hearing loss and music

We will make announcements and guide you through the research activities during the concert.

The study has been given ethical clearance by the University of Leeds (Ref: AHC FREC 4477).



1. BACKGROUND QUESTIONNAIRE

IMPORTANT: By completing this research activity and handing in your booklet, you are giving your consent to these responses being used anonymously for purposes of research publication and for re-use by other researchers via data repositories.

About You

Please complete the following information about you:

1. Seat row and number (*we need this when analysing your answers*).

   Seat Row _____ Seat Number _____

2. Age (years): _____

3. Gender (please circle): Male Female Non-binary/other

4. Are you familiar with Beethoven's 6th Symphony (Pastoral Symphony)?

Please circle a number from 1-10												
Familiarity	Not at all familiar	1	2	3	4	5	6	7	8	9	10	Extremely familiar

5. If you know the work, how much do you like Beethoven's 6th Symphony?

	Please circle a number from 0-10												
Liking	Not at all	0	1	2	3	4	5	6	7	8	9	10	Like very much

neutral

6. Are you a musician?

Please circle which title best describes you:					
Non-musician	Music-loving non-musician	Amateur musician	Serious amateur musician	Semi-professional musician	Professional musician

Please complete the following information about your hearing:

7. Without hearing assistance (hearing aids, implants) how much difficulty do you have in a conversation with one other person in a quiet place?

Please circle which best describes your experience:				
No difficulty	Only slight difficulty	Moderate difficulty	Great difficulty	Cannot manage at all

8. Please indicate (circle) whether you have the following hearing conditions:

Hearing loss (whether one or two ears)	Yes	No	Not sure
Hyperacusis (heightened sensitivity to loud sounds)	Yes	No	Not sure
Tinnitus (extended ringing in the ears when no external sound is present)	Yes	No	Not sure

9. Do you wear hearing aid(s) or cochlear implant(s) devices? Yes / No

   If yes, please indicate with an X what you are currently wearing:

	Left Ear	Right Ear
Nothing		
Hearing Aid		
Cochlear Implant		
Bone-anchored Aid		

10. If you are wearing hearing aids, do you currently have it set to a music program?

    Please circle: Yes / No / Don't know

11. If you are wearing hearing aid(s) or impant(s), do you normally wear them to orchestral concerts?

    Please circle: Never / Sometimes / Always

DEFINITIONS

Here are the attributes we will ask you to rate each of the extracts on. Please take a couple of minutes to familiarise yourself with these. A brief version of these definitions will be also provided next to the questions.

Clarity

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. Unclear music may sound indistinct, mushy or muddy. Clear music may sound clean and distinct.

Distortion

Distortion can be caused by artefacts that should not be there or do not feel right, e.g., noise, hiss, pops or crackles. It can also be caused by pitches sounding wrong. Music with no distortion sounds like an authentic version of what was performed.

Frequency balance

Frequency balance refers to the perceived balance between treble (or higher pitch) and bass (or lower pitch) sounds. Within the research activity, we will ask you to rate whether you are experiencing ‘too much treble’ or ‘too much bass’.

Harshness

Harshness refers to an emphasis of certain sound qualities (often in the treble frequencies or higher pitches) that can feel overwhelming, abrasive, painful, or discomforting. Harsh sounds may sound piercing, screechy, shrill or sharp.

Loudness

Loudness is the perceived “strength” of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is ‘too quiet’ or ‘too loud’

Enjoyment

Music enjoyment is the pleasure, satisfaction, or positive feelings you experience while listening to the music. It may include appreciation of musical beauty, expressiveness, structure, creativity, or artistic value.

3. RESEARCH ACTIVITY

You will hear the orchestra perform 10 music excerpts. They are arranged in pairs, where the orchestra play the same bit of the symphony but in different ways. After each excerpt, please give a score i.e. you will be giving scores 10 times.

We want your honest, individual score for the overall impression as if you were in a normal concert. Please **do not** try and analyse how the orchestra is playing to figure out the "right" answer. There are no wrong or right answers.

The first two excerpts are trial runs so you can get familiar with doing the task.

In each response, you will be required to circle ratings scales. For example:

How much did you enjoy the excerpt (circle a number 1-10)												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	Extremely enjoyable

If you moderately enjoyed listening to the excerpt, you might circle 6, e.g.

How much did you enjoy the excerpt (circle a number 1-10)												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	(6)	7	8	9	10	Extremely enjoyable

You will also be asked to write a sentence or two about your experience:

In 1-2 sentences, what was your overall impression of the musical excerpt:

There will be a pause after each musical excerpt for a period of 1 minute for you to complete your responses. A timer will be displayed on the big screen during this time.

If you make a mistake, please scribble through the circle and/or place an X to indicate this is incorrect.

Please write as clearly as possible so we can accurately record your responses.

Excerpt 1

Please score on **loudness**

Loudness is the perceived “strength” of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is ‘too quiet’ or ‘too loud’

Please circle a number from 1-10 for each row												
How often too quiet	Never	1	2	3	4	5	6	7	8	9	10	Always
How often too loud	Never	1	2	3	4	5	6	7	8	9	10	Always

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	Extremely enjoyable

Excerpt 2

Please score on **loudness**

Loudness is the perceived “strength” of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is ‘too quiet’ or ‘too loud’.

Please circle a number from 1-10 for each row												
How often too quiet	Never	1	2	3	4	5	6	7	8	9	10	Always
How often too loud	Never	1	2	3	4	5	6	7	8	9	10	Always

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	**Extremely enjoyable**

Excerpt 3

Please score on **clarity**

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. **Unclear** music may sound indistinct, mushy, or muddy, **clear music** may sound clean and distinct.

Please circle a number from 1-10												
Clarity	Very unclear	1	2	3	4	5	6	7	8	9	10	Very clear

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	**Extremely enjoyable**

Excerpt 4

Please score on **clarity**

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. **Unclear** music may sound indistinct, mushy, or muddy, **clear music** may sound clean and distinct.

Please circle a number from 1-10												
Clarity	Very unclear	1	2	3	4	5	6	7	8	9	10	Very clear

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	Extremely enjoyable

Excerpt 5

Please score on **loudness** and **distortion**

Loudness is the perceived “strength” of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is ‘too quiet’ or ‘too loud’.

Please circle a number from 1-10 for each row												
How often too quiet	Never	1	2	3	4	5	6	7	8	9	10	Always
How often too loud	Never	1	2	3	4	5	6	7	8	9	10	Always

Distortion can be caused by artefacts that should not be there or do not feel right, e.g., noise, hiss, pops or crackles. It can also be caused by pitches sounding wrong. Music with no distortion sounds like an authentic version of what was performed.

Please circle a number from 1-10												
Distortion	Very unclear	1	2	3	4	5	6	7	8	9	10	Very clear

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	Extremely enjoyable

Excerpt 6

Please score on **loudness** and **distortion**

Loudness is the perceived “strength” of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is ‘too quiet’ or ‘too loud’.

Please circle a number from 1-10 for each row												
How often too quiet	Never	1	2	3	4	5	6	7	8	9	10	Always
How often too loud	Never	1	2	3	4	5	6	7	8	9	10	Always

Distortion can be caused by artefacts that should not be there or do not feel right, e.g., noise, hiss, pops or crackles. It can also be caused by pitches sounding wrong. Music with no distortion sounds like an authentic version of what was performed.

Please circle a number from 1-10												
Distortion	Very unclear	1	2	3	4	5	6	7	8	9	10	Very clear

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	Extremely enjoyable

Excerpt 7

Please score on **frequency balance** and **harshness**

Frequency balance refers to the perceived balance between treble (or higher pitch) and bass (or lower pitch) sounds. Within the research activity, we will ask you to rate whether you are experiencing ‘too much treble’ or ‘too much bass’.

Please circle a number from 1-10 for each row												
Treble	Too little treble	1	2	3	4	5	6	7	8	9	10	Too much treble
Bass	Too little bass	1	2	3	4	5	6	7	8	9	10	Too much bass

Harshness refers to an emphasis of certain sound qualities (often in the treble frequencies or higher pitches) that can feel overwhelming, abrasive, painful, or discomforting. Harsh sounds may sound piercing, screechy, shrill or sharp.

Please circle a number from 1-10												
Harshness	Not at all harsh	1	2	3	4	5	6	7	8	9	10	Very harsh

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	Extremely enjoyable

Excerpt 8

Please score on **frequency balance** and **harshness**

Frequency balance refers to the perceived balance between treble (or higher pitch) and bass (or lower pitch) sounds. Within the research activity, we will ask you to rate whether you are experiencing ‘too much treble’ or ‘too much bass’.

Please circle a number from 1-10 for each row												
Treble	Too little treble	1	2	3	4	5	6	7	8	9	10	Too much treble
Bass	Too little bass	1	2	3	4	5	6	7	8	9	10	Too much bass

Harshness refers to an emphasis of certain sound qualities (often in the treble frequencies or higher pitches) that can feel overwhelming, abrasive, painful, or discomforting. Harsh sounds may sound piercing, screechy, shrill or sharp.

Please circle a number from 1-10												
Harshness	Not at all harsh	1	2	3	4	5	6	7	8	9	10	Very harsh

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	Extremely enjoyable

Excerpt 9

Please score on **Clarity**

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. **Unclear** music may sound indistinct, mushy, or muddy, **clear music** may sound clean and distinct.

Please circle a number from 1-10												
Clarity	Very unclear	1	2	3	4	5	6	7	8	9	10	Very clear

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	**Extremely enjoyable**

Excerpt 10

Please score on **clarity**

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. **Unclear** music may sound indistinct, mushy, or muddy, **clear music** may sound clean and distinct.

Please circle a number from 1-10												
Clarity	Very unclear	1	2	3	4	5	6	7	8	9	10	Very clear

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt?												
Overall Enjoyment	Not at all enjoyable	1	2	3	4	5	6	7	8	9	10	Extremely enjoyable

4. CUE MARKING TASK

The orchestra will now play a full movement of the Symphony.

The cue marking task gives an opportunity for you to indicate any point in the music where you have a noteworthy experience. This may include positive aspects (e.g. a moment you particularly enjoyed) or any perceived issues (e.g. passages too quiet, distortion).

This exercise is optional.

Use the “Time” column to mark the time as displayed on the screen timer (roughly). Then describe your experience in very brief words.

Time (MM:SS)	Experience
_ _ : _ _	

Time (MM:SS)	Experience

**********END OF QUESTIONNAIRE**********